CAN GRAPH STRUCTURE BEAT THE CROWD? PREDICTING CRYPTO PREDICTION MARKET OUTCOMES VIA DYNAMIC MARKET SIMILARITY NETWORKS AND NODE EMBEDDINGS

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ABSTRACT

Prediction markets aggregate collective beliefs into probability estimates, yet standard calibration analyses treat each market independently, ignoring rich relational structure among co-settling contracts. We propose constructing dynamic similarity networks over $\sim$8,800 resolved Polymarket crypto binary contracts and learning node embeddings (via node2vec) that capture each market's structural position. Using a leakage-safe evaluation protocol that groups same-day settlement cohorts, we test whether graph-derived features—combined with market microstructure and options-implied features—improve probabilistic prediction of market outcomes beyond the crowd-implied price, as measured by Brier score, ECE, and log-loss.

1 MOTIVATION AND PROBLEM DEFINITION

Prediction markets are platforms where participants trade binary outcome contracts whose prices aggregate collective beliefs into probability estimates. Platforms such as Polymarket host thousands of daily contracts of the form *"Will BTC be above $X on date D?"*, each resolving to 1 (Yes) or 0 (No) based on the observed spot price at settlement. A fundamental question in market microstructure research is whether market-implied probabilities are *well-calibrated* as in the contract trading at price p truly resolves Yes approximately p fraction of the time (Wolfers & Zitzewitz, 2004).

The standard approach to evaluating prediction market accuracy treats each market as an independent observation: one computes the Brier score or Expected Calibration Error (ECE) of the market price against the realized outcome. However, these markets are far from independent. On any given day, approximately 25 contracts settle on the same underlying asset (BTC or ETH), with strikes spanning a range around the current spot price. These contracts share a common underlying process, have correlated price dynamics, and are connected through shared liquidity providers and informed traders. *This rich relational structure is entirely ignored by standard calibration analysis.*

Network science provides a natural framework for modeling these relationships. In the financial networks literature, correlation-based graphs have been used extensively to study financial markets (Mantegna, 1999; Boginski et al., 2005; Onnela et al., 2003), revealing community structure that mirrors sector classifications, centrality patterns that identify systemically important stocks, and temporal dynamics that detect regime shifts during crises (Isogai, 2017; Fenn et al., 2012). More recently, graph neural networks (GNNs) and node embedding methods have been applied to financial forecasting, demonstrating that network position carries predictive information beyond individual asset features (Zhou et al., 2025; Chen et al., 2018).

Problem Statement. Given a set of $N \approx 8{,}800$ resolved Polymarket crypto markets with top-of-book quotes and price history, we construct dynamic similarity networks over these markets and learn node embeddings that capture each market's structural position. We then evaluate whether these graph-derived features improve *probabilistic* prediction of market resolution outcomes (mea-

sured by Brier score, ECE, and log-loss) compared to the market's own price which is the baseline probability implied by the crowd.

In other words: *Can the structure of the market similarity graph tell us something the crowd doesn't know?*

A critical design constraint: avoiding data leakage. Because ∼25 markets settle daily on the same asset, observing the outcome of any one contract effectively reveals the spot price and thus the outcome of all other same-day contracts. This means (i) all markets in a daily settlement cohort must be grouped together in train or test splits, and (ii) the similarity graph must be constructed from *residualized* returns that strip out the dominant spot-price factor, so that the graph captures genuine structural relationships rather than trivially encoding proximity to the spot price. We describe our approach to these challenges in Section 2.

Related Work. Correlation-based financial networks, pioneered by Mantegna (1999) using minimum spanning trees of equity return correlations, have been extended to dynamic settings where correlation structure reorganizes during market stress (Onnela et al., 2003; Isogai, 2017). Node embedding methods such as node2vec (Grover & Leskovec, 2016) and graph convolutional networks (Kipf & Welling, 2017) have since been applied to financial forecasting—Chen et al. (2018) used GCNs for stock price prediction and Zhou et al. (2025) proposed an evolving multiscale GNN for cryptocurrency volatility. However, none of these works address prediction markets, where the graph structure arises from families of related binary contracts rather than distinct assets.

2 Data Description

Our dataset is sourced from Polymarket, a decentralized prediction market platform, and has **already been fully collected and stored** in a DuckDB database (`crypto_markets.duckdb`, ∼16.5 GB). The data covers the period from August 9, 2025 to March 12, 2026 (∼7 months). The database contains the following tables:

Table 1: Summary of the Polymarket dataset. All data has been collected and stored.

Table	Rows	Description	Usage in project
`book_levels`	308M	Best bid/ask snapshots (top-of-book price and size)	Mid-price computation, liquidity features
`metrics`	82M	Per-market volume, liquidity, and spread over time	Node features (spread, volume, liquidity)
`markets`	8.8K	Market metadata: asset, direction, thresholds, resolution outcome	Ground-truth labels, metadata features
`prices`	353K	Direct price observations	Price return series

The markets covered are daily crypto binary contracts on BTC and ETH with the following types: *above*, *below*, *between*, *reach*, and *dip to X*. Each market has a defined settlement date and resolves to 0 or 1 based on whether the condition was met. Figure 1 below provides an overview of the dataset.

Additionally, we have access to a Deribit options/derivatives database (`backtest_5m.duckdb`, ∼7.4 GB, April 2025 – February 2026) containing 72.6M options snapshots with implied volatility data, futures prices, and computed volatility indices. From this database, we will extract supplementary node features.

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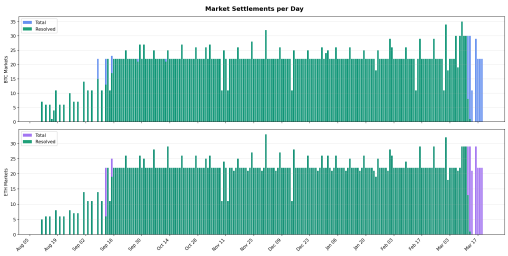
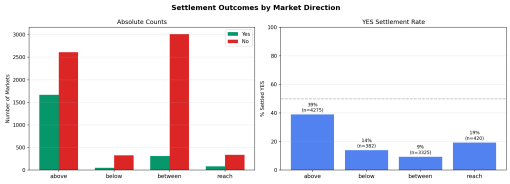
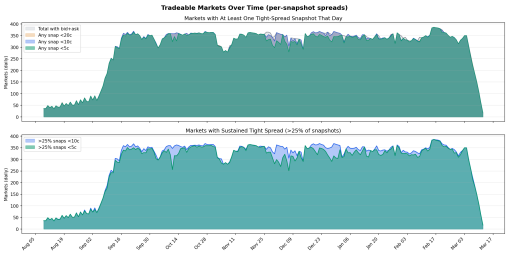
Y. Chen, Z. Wei, and X. Huang. Incorporating corporation relationship via graph convolutional neural networks for stock price prediction. In *CIKM*, pp. 1655–1658, 2018.



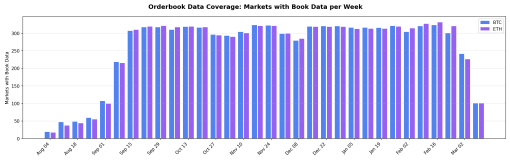
(a) Daily market settlements



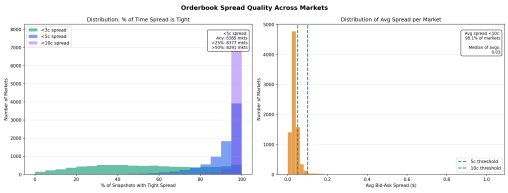
(b) Outcome distribution



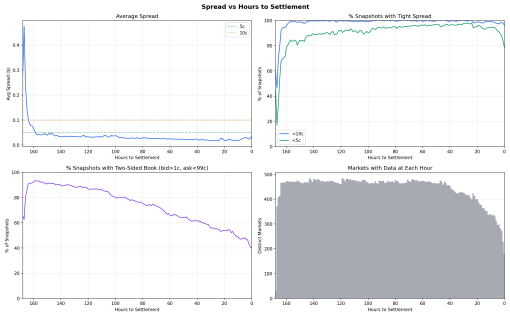
(c) Tradeable markets over time



(d) Orderbook coverage by week



(e) Bid-ask spread distribution



(f) Spread near settlement

Figure 1: Overview of the Polymarket dataset (Aug 2025 – Mar 2026).

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